
Force-Optimized Spectral Descriptors for Quadratic SNAP: A Controlled Study of Compression, Data Efficiency, and Transfer

Anonymous authors

Abstract

Can a label-trained atomic descriptor improve the data efficiency of quadratic SNAP without changing its regression head or sacrificing exact symmetry? We introduce the force-optimized spectral descriptor (FOSD), a fold-specific, 40-dimensional contraction of the 55 invariant $J_{\max} = 4$ bispectrum components. The contraction is learned from an included-only energy–force surrogate, after which the prescribed linear-plus-quadratic least-squares head is refit. This construction is exactly invariant and conservative, but it is also algebraically a rank-40 restriction of quadratic SNAP. On a pre-registered Ta 80/20 split, FOSD reduces held-out energy and force MAE by 58.9% and 12.7%; over all ten 80% draws the reductions are 45.2% and 5.8%. These gains are not full-fit parity: on all Ta data, energy and force MAE are 26.0% and 56.5% higher, and the learned model never reaches the reimplemented full-fit energy error by 95% included data. The limitation strengthens under transfer. Across complete 5–95% learning curves for Mo, Nb, and HCP Re, FOSD loses both metrics in every 80% draw and worsens full-fit energy by 5.9–24.5%. Across 9,600 held-out transformation predictions, FOSD’s worst relative energy and scale-normalized force drifts are 3.65×10^{-14} and 2.27×10^{-14} , respectively. Matched molecular-dynamics costs remain indistinguishable from quadratic SNAP. The controlled result is therefore not a universal learned-descriptor improvement: supervised low-rank compression can regularize sparse in-domain fits, but retaining nearly all surrogate coefficient energy neither guarantees full-data fidelity nor transfer.

1 Introduction

Atomic machine-learning potentials are often divided into a representation of each local environment and a regression from that representation to atomic energy. In SNAP, the representation is a vector of bispectrum invariants of a neighbor density [1]; in its quadratic extension, a linear-plus-quadratic map of those invariants substantially improves the tantalum fit [2]. This separation is attractive scientifically: one can change the descriptor while holding the energy head, loss, data, and forces fixed.

The statistical tension is equally clear. At $J_{\max} = 4$, quadratic SNAP has 55 linear and 1,540 quadratic coefficients. Its flexibility lowers full-training error but produces unstable learning curves when only a small fraction of configurations is available [2]. A task-adapted descriptor could suppress irrelevant directions before the quadratic expansion. Conversely, compression can discard weak directions that matter for derivatives or for structural classes absent from the included fold. It can also create the appearance of efficiency merely by imposing a low-rank regularizer.

This paper isolates that tradeoff. We construct FOSD, a supervised spectral contraction of the invariant bispectrum bank, and retain the exact quadratic-SNAP head and analytic force relation. Descrip-

tor weights are learned anew from the included configurations of every outer fold, so no held-out geometry, energy, force, or validation statistic enters training. The same 40-channel architecture is evaluated on the complete 19-fraction-by-10-draw protocol for Ta, Mo, Nb, and Re: 760 held-out folds, 1,520 held-out model fits, and eight full-data fits. We additionally run the prescribed transformation audit and a matched LAMMPS timing experiment.

The central result is a boundary, not a blanket win. FOSD improves both metrics on the fixed Ta 80/20 split and improves the ten-draw mean, especially by limiting volume-group extrapolation. It does not match the paired full-data fit, does not reach the strict Ta energy threshold by 95% data, and regresses on all three transfer elements. Because the learned descriptor is a linear contraction of invariant primitives, its transformation drift remains at machine precision; because those primitives are still computed, its molecular-dynamics cost is unchanged. These outcomes distinguish in-domain regularization, representation replacement, and evaluator acceleration—three effects that are easy to conflate.

2 Relation to learned atomic representations

Fixed invariant bases. SNAP, SOAP, moment tensor potentials, and the atomic cluster expansion build rotation- and permutation-invariant scalars from local environments [1, 3–6]. Such invariance does not guarantee injectivity at finite body order [7], but it removes symmetry learning from the regression problem. FOSD begins from this invariant setting and learns a label-conditioned subspace; it is not a coordinate network that must discover rotations from augmentation.

Data-adapted descriptors. Proper orthogonal descriptors (POD) construct compact invariant bases from parameterized two- and three-body functions and have been extended to fast many-body and quadratic forms [8–10]. Descriptor parameters and fingerprint subsets have also been optimized or selected from data [11, 12]; learned radial subspaces and compressed neighborhood representations provide related reductions [13, 14]. FOSD therefore does not claim the first learned, optimized, or quadratic compact descriptor. Its narrower contribution is a controlled, leakage-free substitution inside the Wood–Thompson quadratic-SNAP protocol: a force-informed task matrix, an unchanged quadratic head, all outer-fold cells, exact transformation tests, and same-architecture refractory transfer.

End-to-end learned representations. DeepPot-SE and SchNet learn symmetry-preserving representations [15, 16]; NequIP, MACE, and Allegro propagate equivariant features and report strong accuracy, efficiency, or scalability [17–19]. These models show that “learned” and “non-invariant” are not synonyms. They are not matched baselines here because their nonlinear readouts and message passing change more than the descriptor. The broader E(3)-equivariant design space connects these models to ACE-like bases and learnable radial functions [20]. Our purpose is complementary: determine what a supervised descriptor contraction alone buys when the downstream energy form is held fixed.

3 Force-optimized spectral descriptor

3.1 Held-fixed energy and force model

For atom i , let $\mathbf{B}_i \in \mathbb{R}^{55}$ be the $J_{\max} = 4$ SNAP bispectrum vector. Quadratic SNAP assigns

$$E_i = \beta^\top \mathbf{B}_i + \frac{1}{2} \mathbf{B}_i^\top \alpha \mathbf{B}_i, \quad \alpha = \alpha^\top, \quad (1)$$

and $E = \sum_i E_i$. We use LAMMPS compute snap with twojmax=8, rfac0=0.99363, bzeroflag=1, and quadraticflag=1. Thus no constant is fit, the isolated-atom descriptor and energy are zero, and forces are the analytic gradient $-\partial E / \partial \mathbf{R}$. The Ta cutoff is the held-fixed 5.594 Å. A common ZBL reference with inner/outer cutoffs 4.0/4.8 Å is subtracted before fitting and restored in prediction.

FOSD replaces the 55-vector exposed to the head by

$$\mathbf{z}_i = \mathbf{V}^\top \mathbf{B}_i, \quad \mathbf{V} \in \mathbb{R}^{55 \times 40}, \quad (2)$$

and fits the same form as Eq. (1) in $\mathbf{z}_i$. Its Jacobian is $\partial \mathbf{z}_i / \partial \mathbf{R} = \mathbf{V}^\top \partial \mathbf{B}_i / \partial \mathbf{R}$; the force rows are therefore derivatives of the same scalar energy, not a separate force model.

3.2 Included-only supervised spectral learning

The descriptor is learned independently in each outer fold. From included atoms only, define the diagonal RMS matrix

$$S_{mm} = \left(\frac{1}{N_{\text{at}}} \sum_{i \in \text{included}} B_{im}^2 \right)^{1/2}, \quad \mathbf{x}_i = \mathbf{S}^{-1} \mathbf{B}_i, \quad (3)$$

with a 10^{-10} numerical floor. We first fit an auxiliary, unrestricted quadratic model in $\mathbf{x}$ to the included energies and analytic force rows. This surrogate uses a relative ridge of 10^{-8} and yields $(\boldsymbol{\beta}_s, \boldsymbol{\alpha}_s)$. We then form the positive semidefinite task matrix

$$\mathbf{T} = \frac{\boldsymbol{\alpha}_s \boldsymbol{\alpha}_s^\top}{\|\boldsymbol{\alpha}_s\|_F^2} + \frac{\boldsymbol{\beta}_s \boldsymbol{\beta}_s^\top}{\|\boldsymbol{\beta}_s\|_2^2}. \quad (4)$$

If $\mathbf{W}_{40}$ contains its 40 leading eigenvectors, the learned map is $\mathbf{V} = \mathbf{S}^{-1} \mathbf{W}_{40}$. Finally, the auxiliary coefficients are discarded and an 860-coefficient linear-plus-quadratic head is refit on the included rows at relative ridge 10^{-10} . On the full folds, the retained task spectrum is 99.891% (Ta), 99.981% (Mo), 99.991% (Nb), and 99.9997% (Re).

For both methods, energy rows receive base weight 100 and force-component rows weight 1. Before these common weights, each included configuration group is scaled by

$$w_g = \sqrt{\bar{n}/n_g}, \quad (5)$$

where n_g is its number of energy-plus-force rows and $\bar{n}$ is the mean across included groups. This gives equal *untyped* squared row mass to every group; it does not claim exact equality after the 100:1 type weights. All columns are RMS-normalized before the ridge solve, whose absolute penalty is the stated relative value times the mean Gram diagonal. The baseline and FOSD use identical folds, targets, weights, and final-head ridge.

3.3 What is and is not compressed

The 40-channel head has $40 + 40(41)/2 = 860$ coefficients, 46.1% fewer than the 1,595-coefficient baseline head. The factorized model also stores $55 \times 40 = 2,200$ projection parameters, for 3,060 learned scalars in total. More importantly, substituting Eq. (2) into the head gives

$$E_i = (\mathbf{V} \boldsymbol{\beta}_z)^\top \mathbf{B}_i + \frac{1}{2} \mathbf{B}_i^\top (\mathbf{V} \boldsymbol{\alpha}_z \mathbf{V}^\top) \mathbf{B}_i, \quad (6)$$

so the original-basis quadratic matrix has rank at most 40. FOSD is therefore a supervised low-rank restriction of quadratic SNAP. It can be collapsed exactly to 1,595 standard qSNAP coefficients for LAMMPS, but it still evaluates all 55 bispectrum primitives. The method compresses the learned interface and hypothesis class; it neither reduces total factorized storage nor removes the bispectrum evaluator.

Invariance. Because $\mathbf{V}$ is a constant scalar mixing of invariant bispectrum components, translation, rotation, and like-species permutation invariance are exact independently of the fitted values. No transformation augmentation is used. This architectural choice deliberately tests whether supervised compression can improve learning curves without trading away the principal statistical advantage of SNAP.

4 Experimental design

4.1 Data and provenance

For Ta we use every supplied JSON configuration in the 12 Wood–Thompson groups. Direct enumeration gives the inventory in Table 1. This archive does *not* reproduce the paper’s groupwise

Table 1: Datasets after direct enumeration and transfer-archive content deduplication. “Rows” are one energy row plus three force rows per atom.

Material	Configs.	Atoms	Groups	Fit rows
Ta	363	4,224	12	13,035
Mo	3,718	66,660	19	203,698
Nb	3,246	56,386	20	172,404
Re	5,029	101,248	26	308,773

count: the staged files contain 12,672 force components, whereas Section V of the article reports 15,297 force components and a group sum of 362 energies [2]. We treat the published error only as an external anchor and return to this discrepancy in Section 7.

Mo, Nb, and Re come from the staged Andolina–Saidi PBE/VASP DeePMD archives [21]. Our converter reads the canonical `set.*` arrays, preserves archive/system/frame provenance, verifies finite values and ragged offsets, and removes exact content duplicates: 9 of 3,727 Mo source frames, none of 3,246 Nb frames, and 391 of 5,420 Re frames. No external or generated configuration is added. Re contains HCP-derived environments and provides the strongest crystal-symmetry shift from BCC Ta/Mo/Nb.

4.2 Outer-fold protocol and metrics

For each material, nominal included fractions are 5%, 10%, ..., 95%, with ten draws per fraction. RNG seed 20260731 is offset by 1,000 times the fraction index and by the draw index. Configurations are shuffled within each group; per-group counts are rounded to the nearest integer and clamped so both sides are nonempty. Every configuration, and hence all of its energy and force rows, lies wholly on one side. Descriptor scaling, auxiliary fitting, cutoff selection, projection learning, and final fitting see only the included side. A separate 100% fit has no held-out rows. Explicit IDs for all folds are saved.

Energy MAE is first averaged over configurations within a group; force MAE is the mean absolute Cartesian-component error, atom weighted within a group. The headline aggregate is the arithmetic mean of group errors. For Ta we also average the four slow-converging groups GSF(110), GSF(112), liquid, and surface. Curves report mean and sample standard deviation over all ten draws. In log-scale figures, a mean-minus-standard-deviation bound that crosses zero is truncated at 5% of the mean for display only; complete numerical mean and standard-deviation tables are unmodified.

We predefine strict data efficiency F^* as the earliest fraction whose mean held-out FOSD energy MAE is no larger than the reimplemented quadratic-SNAP *full included-fit* error. Because this mixes held-out and training residuals, it is deliberately stringent. We also report a budget-matched secondary threshold using quadratic SNAP’s mean held-out 95% error. The same definitions are applied to every material.

4.3 Two baselines and numerical implementation

The literature anchor is Wood and Thompson’s Supplemental Table SII, $J_{\max} = 4$, $K = 55$, $r_{\text{cut}} = 5.594 \text{ \AA}$, and energy MAE 0.00898 eV/atom [2]. The paired baseline uses LAMMPS’s exact 1,595-column SNAP energy/derivative matrix and our deterministic group-balanced ridge in every fold. As an independent pipeline check, the supplied FitSNAP configuration and SVD solver give pooled full-archive MAE 0.000238498 eV/atom and 0.015850 eV/Å; the equal-group paired solver gives 0.000347496 and 0.0133098. We do not relabel the pooled FitSNAP statistic as the equal-group metric.

For transfer, the same architecture is retrained, including material-specific projection weights. A label-free cutoff is selected inside each fold:

$$r_{\text{cut}} = \text{clip}_{[5,6]} \left(\text{round}_{0.1} \left[2q_{0.90}(d_{\text{NN}}^{\text{included}}) \right] \right) \text{ \AA}. \quad (7)$$

Candidate descriptor caches are 5.3/5.4/5.5 Å (Mo), 5.7/5.8 Å (Nb), and 5.2/5.3 Å (Re); full fits select 5.4, 5.7, and 5.3 Å. Both methods use the same selected cutoff in a fold. Transfer ridge solves

Table 2: Headline accuracy, data efficiency, and cost. Full-data columns are included-set fits; 80% columns are mean $\pm$ s.d. over all ten outer draws. F^* is the earliest held-out fraction whose mean energy MAE reaches the corresponding reimplemented full-data fit; $> 95\%$ means it did not. The external paper row uses Wood and Thompson, Supplemental Table SII, $J_{\max} = 4$; that source reports only full-fit energy, while its Figure 5 is GSF(110)-only, so a global force value, matched 80% result, and F^* are not reported or defined. Its unit cost denotes the same $J_{\max} = 4$ quadratic-SNAP evaluator. Other costs are measured within material on the present CPU.

Method	Mat.	Full δE (eV/atom)	Full δF (eV/Å)	80% δE (eV/atom)	80% δF (eV/Å)	F^*	Cost rel.
qSNAP (paper)	Ta	0.00898	not reported	not reported	not reported	not defined	1.0000
qSNAP (paired)	Ta	0.00035	0.01331	0.7691 ± 1.3537	0.0604 ± 0.0133	reference	1.0000
qSNAP (paired)	Mo	0.03087	0.52083	0.0337 ± 0.0067	0.6076 ± 0.0669	reference	1.0000
qSNAP (paired)	Nb	0.00371	0.10756	0.0037 ± 0.0010	0.1130 ± 0.0049	reference	1.0000
qSNAP (paired)	Re	0.01795	0.80672	0.0211 ± 0.0069	0.8370 ± 0.0162	reference	1.0000
FOSD	Ta	0.00044	0.02084	0.4213 ± 0.6197	0.0569 ± 0.0122	all/slow $> 95\%/> 95\%$	0.9987
FOSD	Mo	0.03271	0.54128	0.0357 ± 0.0061	0.6365 ± 0.0783	95%	1.0066
FOSD	Nb	0.00425	0.12735	0.0042 ± 0.0010	0.1309 ± 0.0057	$> 95\%$	1.0003
FOSD	Re	0.02234	0.90651	0.0248 ± 0.0077	0.9119 ± 0.0335	$> 95\%$	0.9999

run in float64 on one NVIDIA H200 with TF32 disabled; comparison against the CPU solver gives maximum absolute prediction differences 1.60×10^{-4} for qSNAP and 3.10×10^{-5} for FOSD over the concatenated design rows.

4.4 Transformation and timing audits

For each material, we fit the pre-registered 80%, draw-0 models with no augmentation and randomly choose ten held-out configurations using a saved, material-specific seed. Each configuration undergoes 100 independent uniform $SO(3)$ rotations, ten translations with each component uniform on $[-1, 1]$ Å, and ten random atom permutations. Rotated reference forces are transformed covariantly before comparison. We retain absolute, component-relative (denominator floor 10^{-12}), vector-relative, and global scale-normalized drift for every sample.

For cost, both draw-0 models are collapsed to standard qSNAP coefficients and run through the same single-rank CPU LAMMPS path with ZBL. The matched 128-atom systems are $4 \times 4 \times 4$ conventional BCC cells for Ta, Mo, and Nb and a $4 \times 4 \times 2$ orthogonal HCP cell for Re. Each starts at 300 K and is integrated for 20 warm-up and 200 timed 1-fs NVE steps. Five repeats use alternating model order and report wall time per atom-step.

5 Results

5.1 Ta: sparse-fold parity, not full-fit parity

The pre-registered 80%, draw-0 split is the operational matched-accuracy test. FOSD obtains 0.008356 eV/atom and 0.059588 eV/Å, versus 0.020344 and 0.068242 for paired qSNAP: improvements of 58.9% and 12.7%. The same ordering holds in the slow-group aggregate, where FOSD gives 0.003972 and 0.144945 versus 0.007451 and 0.176509. Thus the descriptor clears both metrics on this fixed split.

The conclusion is weaker over all draws and reverses at full fit. At 80%, FOSD’s all-group mean is 0.4213 ± 0.6197 eV/atom and 0.0569 ± 0.0122 eV/Å, compared with 0.7691 ± 1.3537 and 0.0604 ± 0.0133 . It wins 8/10 paired energy draws and 7/10 force draws. For the slow groups, the corresponding means are 0.01213 ± 0.01664 and 0.13637 ± 0.03756 , versus 0.01366 ± 0.02167 and 0.15428 ± 0.03994 ; wins are only 5/10 for energy but 9/10 for force. At 100% included data, FOSD is worse: 0.000438/0.020836 versus 0.000347/0.013310. The 80% gain is therefore regularization under holdout, not a uniformly more expressive full-data representation.

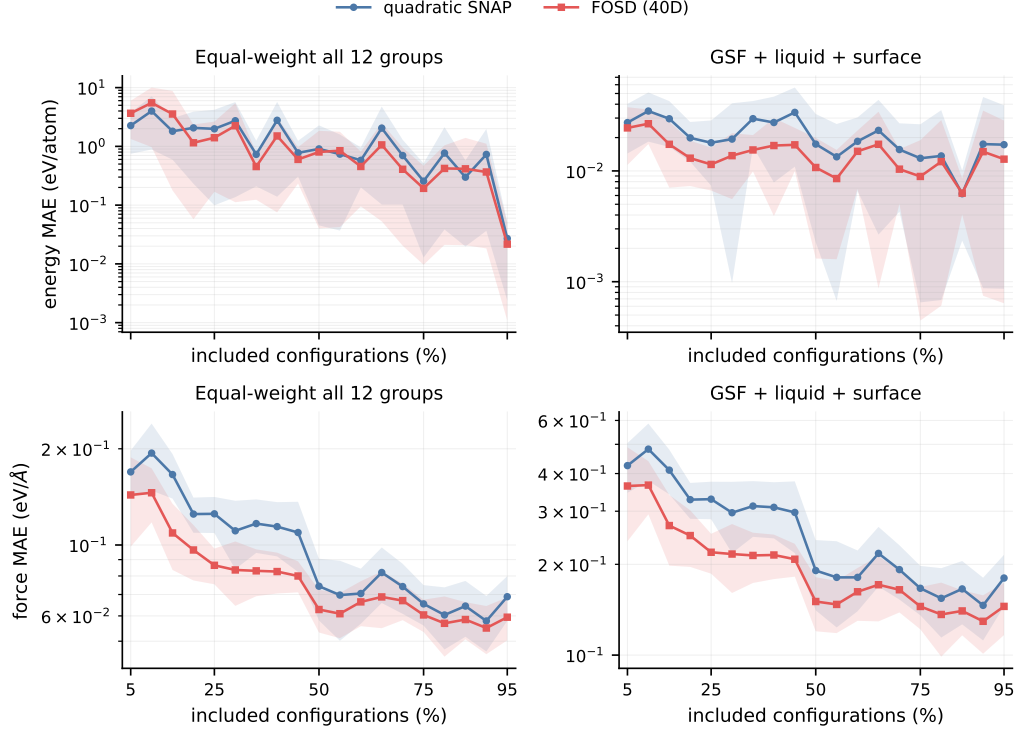


Figure 1: Ta held-out learning curves. Lines are ten-draw means and shading is one standard deviation; a nonpositive lower display bound is truncated as described in Section 4. Equal-group energy is dominated by sparse volume-curve extrapolation, whereas the slow-group force curve shows a more consistent FOSD advantage.

Table 3: Earliest FOSD fraction F^* under two consistent references. The Ta slow subset is reported separately. $> 95\%$ means the mean held-out curve did not reach the reference anywhere on the measured grid.

Material	Aggregate	qSNAP full fit	qSNAP 95% held out
Ta	all groups	$> 95\%$	95%
Ta	GSF+liquid+surface	$> 95\%$	20%
Mo	all groups	95%	$> 95\%$
Nb	all groups	$> 95\%$	55%
Re	all groups	$> 95\%$	$> 95\%$

5.2 Data-efficiency thresholds

Figure 1 contains every nominal fraction. Neither the all-group nor slow-group FOSD curve reaches the strict paired full-fit Ta energy reference by 95% data. Under the secondary, held-out qSNAP-at-95% reference, F^* is 95% for all groups and 20% for the slow subset. Against the published 0.00898 external anchor, the slow subset reaches 55%; the all-group curve does not. These three statements are intentionally kept separate: only the first is the strict primary definition.

The apparent Ta all-group scale in Fig. 1 is not a plotting artifact. At 80%, mean Volume-BCC energy MAE is 8.761 eV/atom for qSNAP and 4.753 for FOSD; Volume-A15 is 0.1305 versus 0.0477, and Volume-FCC is 0.2814 versus 0.2045. Their symmetry-forced force errors remain near 10^{-14} eV/Å. Low rank limits some energy blow-ups but does not make them acceptable. Within the slow subset, the mean force gain comes from liquid (0.4588 to 0.4017) and surface (0.1321 to 0.1079); FOSD is worse on GSF(110) (0.00836 to 0.01201) and GSF(112) (0.01788 to 0.02388).

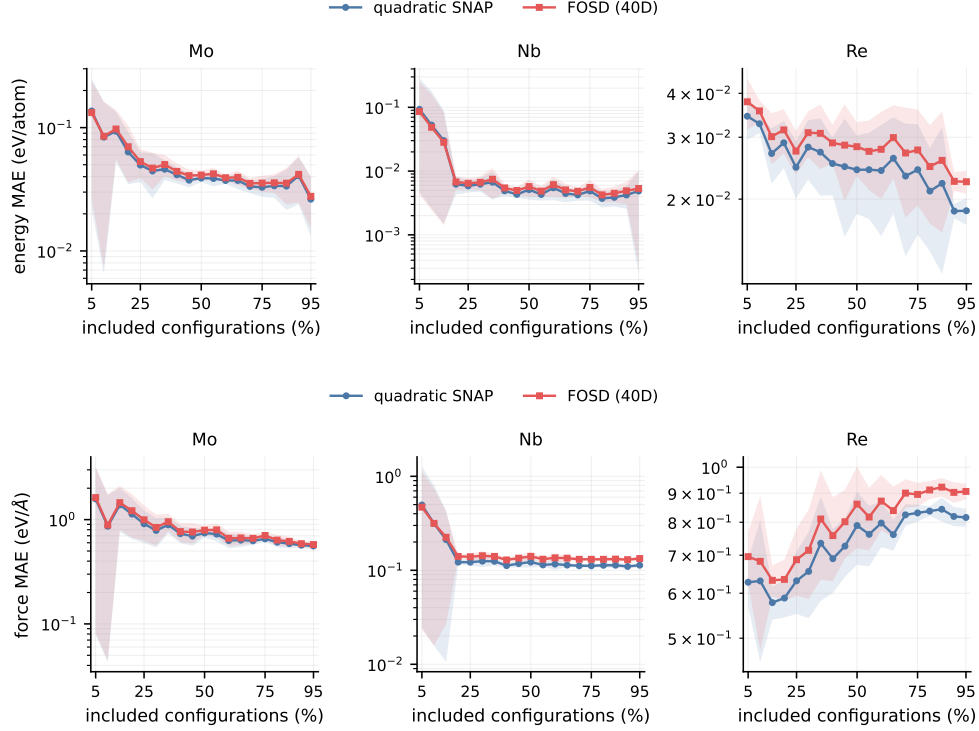


Figure 2: Equal-group held-out transfer curves for Mo, Nb, and HCP Re. Means and standard deviations use all ten draws. The architecture and dimension are unchanged; weights and the label-free cutoff are learned inside each fold. FOSD is consistently above the baseline, most clearly for HCP Re.

This groupwise decomposition explains why the subset average improves without a universal defect-physics improvement.

5.3 Transfer: compression does not generalize

Transfer removes the ambiguity in Ta’s high-variance energy aggregate. At 80%, FOSD loses both energy and force in all ten paired draws for each of Mo, Nb, and Re. The mean pairs (FOSD versus qSNAP) are 0.03571/0.63651 versus 0.03373/0.60763 for Mo, 0.004245/0.13091 versus 0.003728/0.11301 for Nb, and 0.02477/0.91188 versus 0.02108/0.83698 for Re. Full-data energy increases by 5.9%, 14.4%, and 24.5%, while force increases by 3.9%, 18.4%, and 12.4%, respectively. Mo alone reaches its baseline full-fit energy at 95%; Nb reaches the secondary qSNAP-95% threshold at 55%, and Re reaches neither. Per-group panels in Appendix A show isolated cells where the rank restriction helps, but the equal-group conclusion is not driven by aggregation: no transfer material exhibits paired 80% wins.

5.4 Exact symmetry, unchanged cost

Table 4 confirms the architectural prediction. Under all four materials, FOSD’s worst energy-relative drift is 3.65×10^{-14} (Re rotation), compared with 4.37×10^{-14} for qSNAP. Its worst force drift normalized by the test’s force scale is 2.27×10^{-14} (Ta rotation), compared with 2.54×10^{-14} for qSNAP. Every per-material translation and permutation result is likewise at floating-point precision. The much larger componentwise ratios, up to 0.0535 for FOSD and 0.0619 for qSNAP, occur for Ta where the reference component itself is at the 10^{-12} denominator floor; the all-material scale normalization shows no physical symmetry failure. Reporting both normalizations prevents near-zero components from being hidden or mistaken for such a failure.

Table 4: Maximum per-material invariance drift over the prescribed held-out audits. Each material uses 1,000 rotations, 100 translations, and 100 permutations per model. The component ratio uses denominator $\max(|F_{ja}|, 10^{-12})$ and is consequently controlled by near-zero force components; the final column normalizes the largest absolute drift by the largest force magnitude in the same test.

Mat.	Method	Test	Samples	$\max \Delta E/E $	max component $ \Delta F/F $	$\max \Delta F /\max F $
Ta	quadratic SNAP	rotation	1000	4.58e-15	6.19e-02	2.54e-14
Ta	quadratic SNAP	translation	100	2.62e-15	3.56e-02	1.36e-14
Ta	quadratic SNAP	permutation	100	1.96e-15	5.39e-03	6.26e-15
Ta	FOSD (40D)	rotation	1000	4.29e-15	5.35e-02	2.27e-14
Ta	FOSD (40D)	translation	100	2.36e-15	2.92e-02	9.49e-15
Ta	FOSD (40D)	permutation	100	1.31e-15	4.67e-03	3.13e-15
Mo	quadratic SNAP	rotation	1000	7.75e-15	1.21e-10	1.88e-14
Mo	quadratic SNAP	translation	100	3.80e-15	1.10e-10	1.05e-14
Mo	quadratic SNAP	permutation	100	5.22e-15	1.07e-10	6.86e-15
Mo	FOSD (40D)	rotation	1000	6.23e-15	4.55e-10	1.03e-14
Mo	FOSD (40D)	translation	100	3.19e-15	1.60e-10	8.67e-15
Mo	FOSD (40D)	permutation	100	3.32e-15	5.52e-11	5.09e-15
Nb	quadratic SNAP	rotation	1000	4.73e-15	2.81e-07	1.57e-14
Nb	quadratic SNAP	translation	100	2.37e-15	1.71e-07	1.35e-14
Nb	quadratic SNAP	permutation	100	1.48e-15	7.26e-08	8.38e-15
Nb	FOSD (40D)	rotation	1000	2.58e-15	1.85e-07	1.76e-14
Nb	FOSD (40D)	translation	100	1.51e-15	1.64e-07	1.20e-14
Nb	FOSD (40D)	permutation	100	1.72e-15	4.16e-08	7.49e-15
Re	quadratic SNAP	rotation	1000	4.37e-14	1.99e-10	7.70e-15
Re	quadratic SNAP	translation	100	2.04e-14	1.60e-10	6.37e-15
Re	quadratic SNAP	permutation	100	1.75e-14	9.87e-11	6.33e-15
Re	FOSD (40D)	rotation	1000	3.65e-14	7.24e-11	8.05e-15
Re	FOSD (40D)	translation	100	2.77e-14	5.93e-11	4.80e-15
Re	FOSD (40D)	permutation	100	1.36e-14	5.07e-11	4.99e-15

Table 5: Matched 128-atom LAMMPS timing (five repeats, 200 timed steps): BCC Ta/Mo/Nb and HCP Re. Relative cost is computed within each material.

Material	Method	s atom ⁻¹ step ⁻¹	Relative
Ta	quadratic SNAP	2.957e-04 $\pm$ 1.3e-06	1.0000
Ta	FOSD (40D)	2.953e-04 $\pm$ 7.8e-07	0.9987
Mo	quadratic SNAP	3.053e-04 $\pm$ 1.1e-06	1.0000
Mo	FOSD (40D)	3.073e-04 $\pm$ 4.6e-07	1.0066
Nb	quadratic SNAP	3.080e-04 $\pm$ 3.9e-07	1.0000
Nb	FOSD (40D)	3.081e-04 $\pm$ 4.9e-07	1.0003
Re	quadratic SNAP	3.152e-04 $\pm$ 2.7e-07	1.0000
Re	FOSD (40D)	3.151e-04 $\pm$ 3.3e-07	0.9999

The timing result is also predicted by Eq. (6). qSNAP takes $(2.957 \pm 0.013) \times 10^{-4}$ s per atom-step on Ta and FOSD $(2.953 \pm 0.008) \times 10^{-4}$, a ratio of 0.9987. The corresponding FOSD/qSNAP ratios are 1.0066 for Mo, 1.0003 for Nb, and 0.9999 for HCP Re. Thus the largest observed departure from unity across all four matched systems is 0.66%. Exact coefficient collapse removes an explicit projection at runtime, but LAMMPS still constructs the same bispectrum and quadratic SNAP basis; indeed, after collapse both timing paths execute the same evaluator with different coefficient values. FOSD is therefore not an inference-speed improvement; a faster result would require a new primitive evaluator or kernel, not coefficient compression alone.

Training overhead is modest only after descriptor matrices are cached. The complete 382-run Ta sweep took 111.3 s with CPU linear algebra; the Mo, Nb, and Re sweeps took 31.9, 29.0, and 44.7 s on the H200. One-time exact SNAP extraction dominates: 25.0 s for Ta and, across every candidate cutoff, 19.6 min for Mo, 9.8 min for Nb, and 20.3 min for Re. These caches are shared by qSNAP and FOSD, so they do not favor either fit, but they reinforce the cost conclusion: learning and solving the projection are cheap compared with evaluating the unchanged primitives.

6 Mechanism, sensitivity, and failure analysis

A low-rank bias explains the Ta pattern. Equation (6) makes the mechanism explicit: FOSD does not add representational directions; it removes 15 directions from the quadratic coefficient matrix. On sparse Ta folds this bias damps poorly constrained volume extrapolation and often improves force generalization. With all data, the missing directions are identifiable and the unrestricted model fits better. On transfer sets, retaining 99.98–99.9997% of the surrogate task spectrum still loses accuracy. Coefficient-norm energy in Eq. (4) does not weight discarded directions by held-out feature covariance or by their spatial Jacobians, so a spectrally small direction can remain predictively important.

Ridge and dimension do not rescue a universal claim. A post-freeze sensitivity study evaluates all ten draws at 50%, 80%, and 95%. At 80%, raising the final ridge from 10^{-10} to 10^{-8} changes FOSD’s all-group mean from 0.4213/0.05688 to 0.1286/0.05272, but qSNAP simultaneously changes from 0.7691/0.06041 to 0.1312/0.04867: FOSD retains a small energy advantage and loses force parity. At 10^{-6} and 10^{-4} , force error degrades further. We retain the pre-frozen 10^{-10} value rather than selecting a ridge on outer-test curves. For dimensions 24, 32, 40, and 48, the 80% FOSD force means are 0.10547, 0.07248, 0.05688, and 0.05631 eV/Å. Forty is the compression elbow; 48 adds 364 head coefficients for a 1.0% force change and worsens mean energy from 0.4213 to 0.7268.

Why retain bispectrum primitives? We attempted genuinely different invariant banks before freezing FOSD. A 24-channel contraction of 31 two-/three-body POD primitives gave 0.06295/0.4705 on the fixed Ta split; 32 channels from 72 two-/three-/four-body POD primitives gave 0.05127/0.5716. A nonlinear student with the 72 POD channels, 128 fixed tanh features, and included-only value-and-Jacobian distillation gave 0.117015/1.66685, with low-symmetry force MAE 4.6895. These failures show that matching local descriptor values is insufficient when the energy gradient depends on out-of-sample descriptor Jacobians. They also motivate the conservative final design, while limiting its novelty: FOSD is supervised invariant compression, not an independent replacement for the internal bispectrum calculation.

7 The published-anchor gap

The published 0.00898 eV/atom value is 25.84 times the paired qSNAP full-fit energy MAE of 0.0003475 eV/atom; the direct FitSNAP SVD check is smaller still. This is not evidence that the 2018 result was wrong, nor is it a favorable baseline adjustment. The staged archive has 2,625 fewer force components than the paper’s inventory, while containing 363 JSON energies against a published group sum of 362. The paired pipeline also uses explicit column normalization, relative ridge, and group balancing rather than the paper’s optimized group weights and historical fitting path. These differences make the external number non-comparable to the paired full fit.

The implication cuts against FOSD: the very low paired training residual makes strict F^* unusually hard to reach, but it also exposes FOSD’s full-data loss directly. We therefore use the reimplementations only for within-pipeline paired claims, report the paper number as a separately sourced anchor, and give both the strict and held-out-95% efficiency thresholds. A future archival reproduction should first reconcile configuration and force inventories before attributing the remaining factor to weighting or solver choices.

8 Limitations and conclusion

FOSD is deliberately narrow. It uses a linear, material-specific contraction of a fixed single-element bispectrum bank; multielement chemical channels are untested. It evaluates only supplied PBE/VASP configurations, not new DFT, long-time stability, phase transitions, elastic constants, or defect formation energies. The transfer datasets differ from Ta in both source and group construction, so their result tests same-architecture generalization rather than zero-shot weight transfer. The strict F^* reference is a training residual and should be read together with the held-out-95% reference. Finally, exact invariance follows from the bispectrum primitives and does not demonstrate that a symmetry-unconstrained descriptor can learn invariance from 363 structures.

Within those limits, the full protocol gives a useful negative constraint on learned descriptor substitution. A force-informed spectral contraction can clear a fixed Ta 80/20 parity test, improve average sparse-fold forces, and retain machine-precision symmetry. It does not improve the full fit, meet strict Ta sample efficiency, transfer to Mo/Nb/Re, or accelerate MD. The learned task spectrum is therefore not by itself a reliable criterion for discarding atomic-feature directions. Future compact descriptors should optimize a training-only objective that explicitly reflects feature covariance and force-Jacobian coverage, and they must reduce the primitive evaluator—not merely the regression head—before claiming computational speed.

Reproducibility statement

Every reported run has a timestamp, seed, fold membership, hyperparameters, included/excluded counts, and groupwise energy/force metrics under logs/. The protocol validator records zero duplicate cells and zero missing values for all four materials. Complete ten-draw summaries are tables/cv_summary_{ta,mo,nb,re}.csv; raw cells are logs/cv_metrics_{ta,mo,nb,re}.csv; explicit configuration IDs are logs/fold_manifest_{ta,mo,nb,re}.jsonl. Published values and their exact source locations are isolated in logs/paper_reference.md. Conversion provenance, invariance samples, timing samples, numerical validation, pilots, and unsuccessful architectures are retained separately. The analysis script refuses incomplete 19-by-10 grids, incomplete transformation cells, non-128-atom timings, or missing audit values before producing any paper table or figure.

References

- [1] A. P. Thompson, L. P. Swiler, C. R. Trott, S. M. Foiles, and G. J. Tucker. Spectral neighbor analysis method for automated generation of quantum-accurate interatomic potentials. *Journal of Computational Physics*, 285:316–330, 2015. <https://doi.org/10.1016/j.jcp.2014.12.018>.
- [2] M. A. Wood and A. P. Thompson. Extending the accuracy of the SNAP interatomic potential form. *The Journal of Chemical Physics*, 148:241721, 2018. <https://doi.org/10.1063/1.5017641>.
- [3] A. P. Bartók, R. Kondor, and G. Csányi. On representing chemical environments. *Physical Review B*, 87:184115, 2013. <https://doi.org/10.1103/PhysRevB.87.184115>.
- [4] A. V. Shapeev. Moment tensor potentials: A class of systematically improvable interatomic potentials. *Multiscale Modeling & Simulation*, 14:1153–1173, 2016. <https://doi.org/10.1137/15M1054183>.
- [5] R. Drautz. Atomic cluster expansion for accurate and transferable interatomic potentials. *Physical Review B*, 99:014104, 2019. <https://doi.org/10.1103/PhysRevB.99.014104>.
- [6] G. Dusson, M. Bachmayr, G. Csányi, R. Drautz, S. Etter, C. van der Oord, and C. Ortner. Atomic cluster expansion: Completeness, efficiency and stability. *Journal of Computational Physics*, 454:110946, 2022. <https://doi.org/10.1016/j.jcp.2022.110946>.
- [7] S. N. Pozdnyakov, M. J. Willatt, A. P. Bartók, C. Ortner, G. Csányi, and M. Ceriotti. Incompleteness of atomic structure representations. *Physical Review Letters*, 125:166001, 2020. <https://doi.org/10.1103/PhysRevLett.125.166001>.
- [8] N. C. Nguyen and A. Rohskopf. Proper orthogonal descriptors for efficient and accurate interatomic potentials. *Journal of Computational Physics*, 480:112030, 2023. <https://doi.org/10.1016/j.jcp.2023.112030>.
- [9] N.-C. Nguyen. Fast proper orthogonal descriptors for many-body interatomic potentials. *Physical Review B*, 107:144103, 2023. <https://doi.org/10.1103/PhysRevB.107.144103>.
- [10] N. C. Nguyen. Proper orthogonal descriptors for multi-element chemical systems. *Journal of Computational Physics*, 511:113102, 2024. <https://doi.org/10.1016/j.jcp.2024.113102>.

- [11] H. Gao, J. Wang, and J. Sun. Improve the performance of machine-learning potentials by optimizing descriptors. *The Journal of Chemical Physics*, 150:244110, 2019. <https://doi.org/10.1063/1.5097293>.
- [12] G. Imbalzano, A. Anelli, D. Giofré, S. Klees, J. Behler, and M. Ceriotti. Automatic selection of atomic fingerprints and reference configurations for machine-learning potentials. *The Journal of Chemical Physics*, 148:241730, 2018. <https://doi.org/10.1063/1.5024611>.
- [13] A. Goscinski, F. Musil, S. Pozdnyakov, J. Nigam, and M. Ceriotti. Optimal radial basis for density-based atomic representations. *The Journal of Chemical Physics*, 155:104106, 2021. <https://doi.org/10.1063/5.0057229>.
- [14] J. P. Darby, J. R. Kermode, and G. Csányi. Compressing local atomic neighbourhood descriptors. *npj Computational Materials*, 8:166, 2022. <https://doi.org/10.1038/s41524-022-00847-y>.
- [15] L. Zhang, J. Han, H. Wang, W. A. Saidi, R. Car, and W. E. End-to-end symmetry preserving inter-atomic potential energy model for finite and extended systems. *Advances in Neural Information Processing Systems*, 31, 2018.
- [16] K. T. Schütt, H. E. Sauceda, P.-J. Kindermans, A. Tkatchenko, and K.-R. Müller. SchNet—A deep learning architecture for molecules and materials. *The Journal of Chemical Physics*, 148:241722, 2018. <https://doi.org/10.1063/1.5019779>.
- [17] S. Batzner, A. Musaelian, L. Sun, M. Geiger, J. P. Mailoa, M. Kornbluth, N. Molinari, T. E. Smidt, and B. Kozinsky. E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials. *Nature Communications*, 13:2453, 2022. <https://doi.org/10.1038/s41467-022-29939-5>.
- [18] I. Batatia, D. P. Kovács, G. N. C. Simm, C. Ortner, and G. Csányi. MACE: Higher order equivariant message passing neural networks for fast and accurate force fields. *Advances in Neural Information Processing Systems*, 35, 2022.
- [19] A. Musaelian, S. Batzner, A. Johansson, L. Sun, C. J. Owen, M. Kornbluth, and B. Kozinsky. Learning local equivariant representations for large-scale atomistic dynamics. *Nature Communications*, 14:579, 2023. <https://doi.org/10.1038/s41467-023-36329-y>.
- [20] I. Batatia, S. Batzner, D. P. Kovács, A. Musaelian, G. N. C. Simm, R. Drautz, C. Ortner, B. Kozinsky, and G. Csányi. The design space of E(3)-equivariant atom-centred interatomic potentials. *Nature Machine Intelligence*, 7:56–67, 2025. <https://doi.org/10.1038/s42256-024-00956-x>.
- [21] C. M. Andolina and W. A. Saidi. Highly transferable atomistic machine-learning potentials from curated and compact datasets across the periodic table. *Digital Discovery*, 2:1070–1082, 2023. <https://doi.org/10.1039/D3DD00046J>.

A All Ta groupwise learning curves

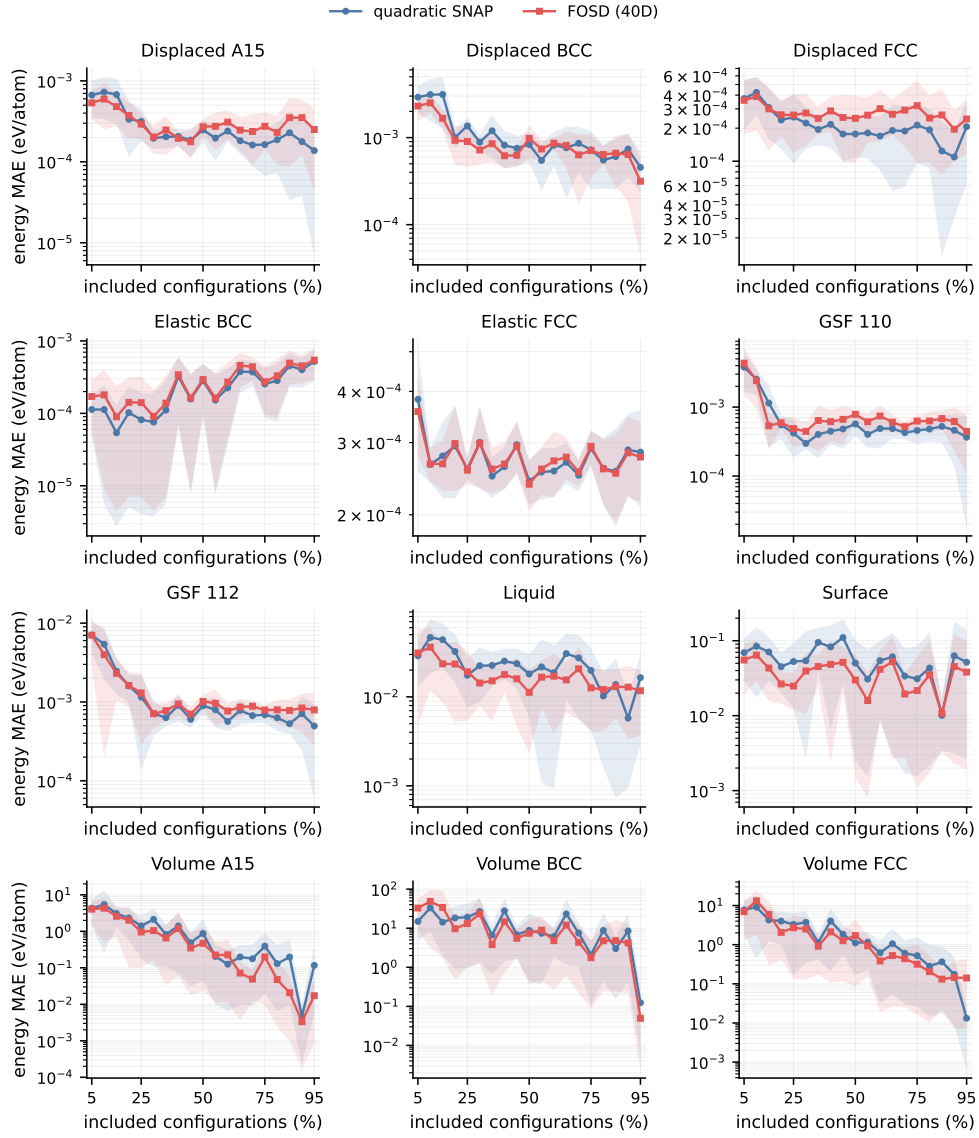


Figure 3: Held-out energy MAE for every Ta group, with all 19 fractions and ten draws. These panels are the source of the aggregate mechanism analysis in Section 5; no group is omitted from the numerical summaries.

B Transfer groupwise learning curves

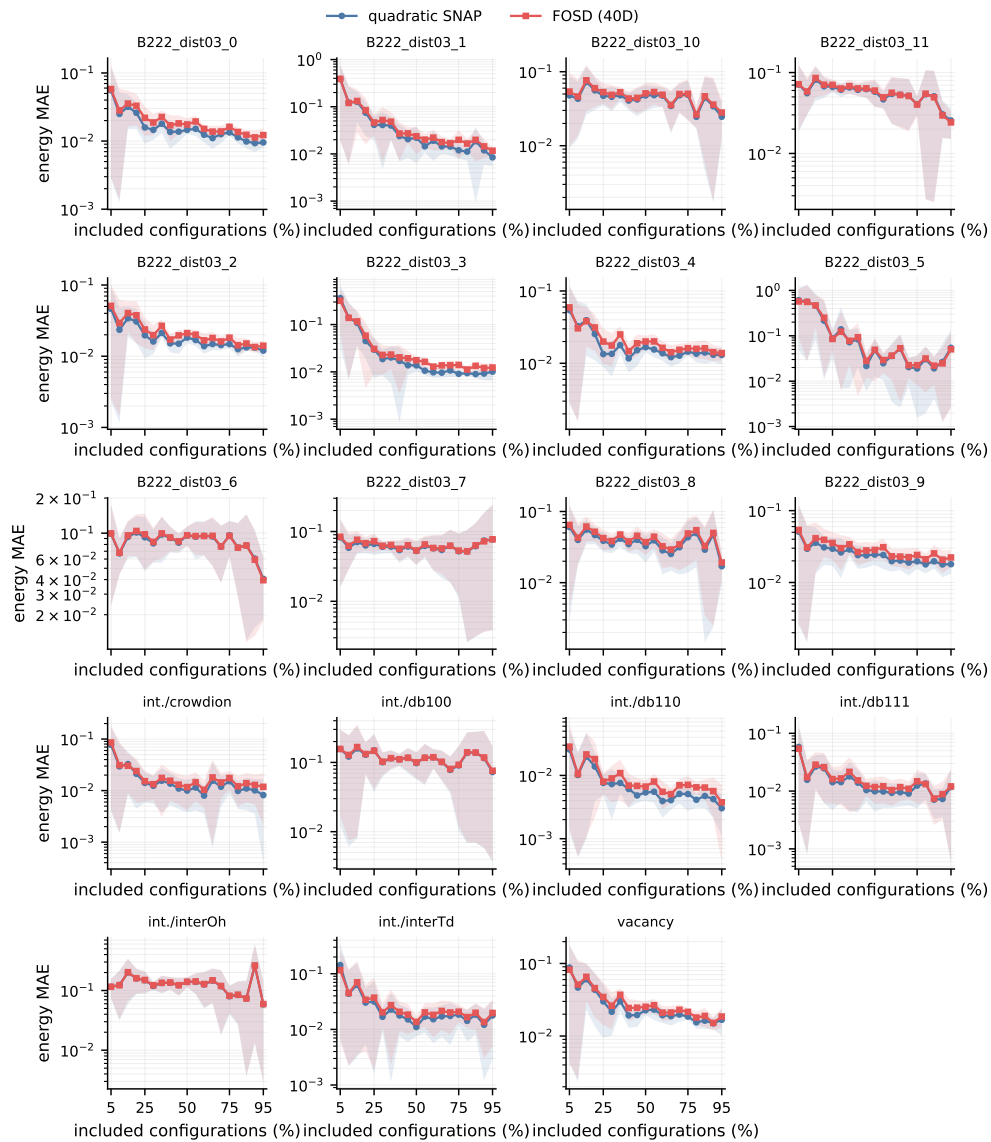


Figure 4: Mo held-out energy MAE for all 19 source-defined groups.

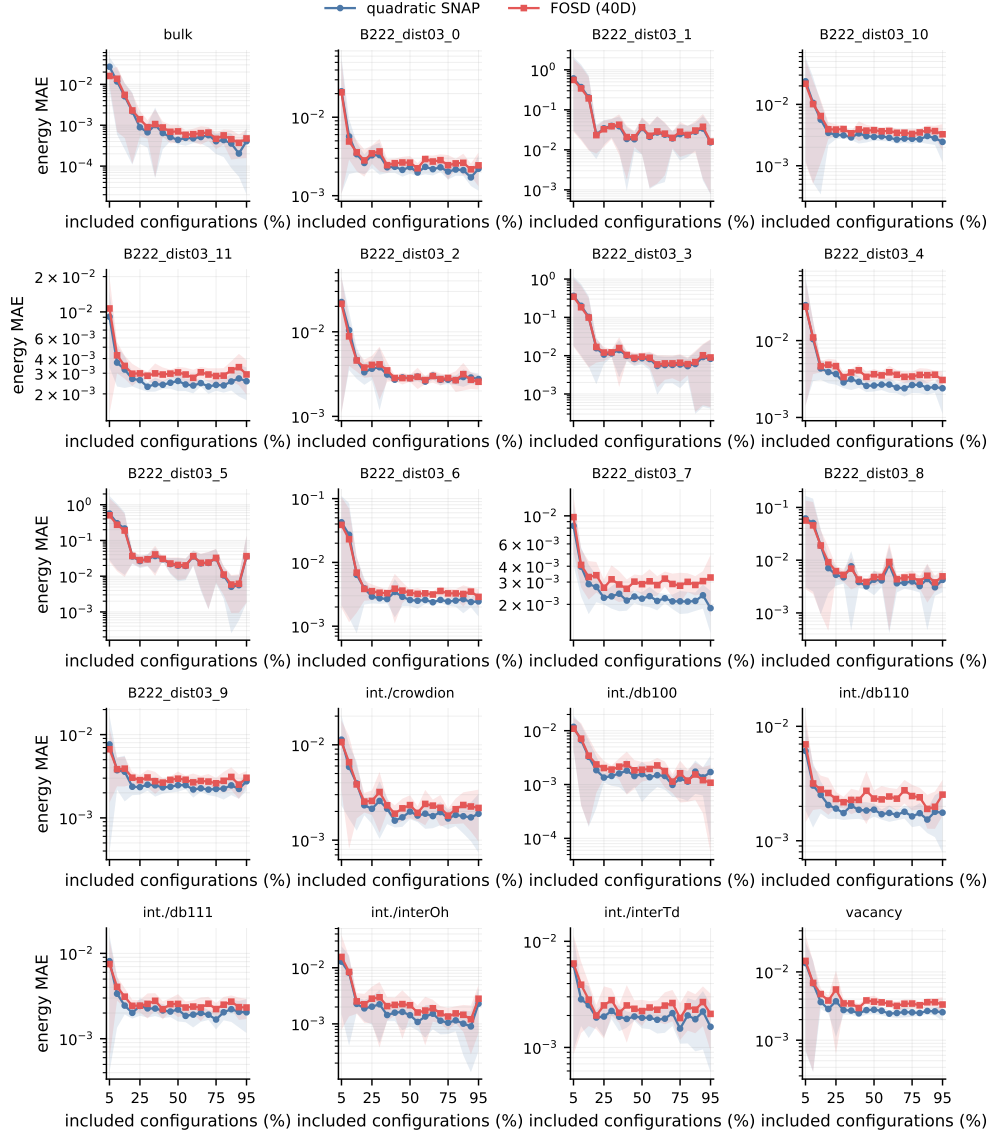


Figure 5: Nb held-out energy MAE for all 20 source-defined groups.

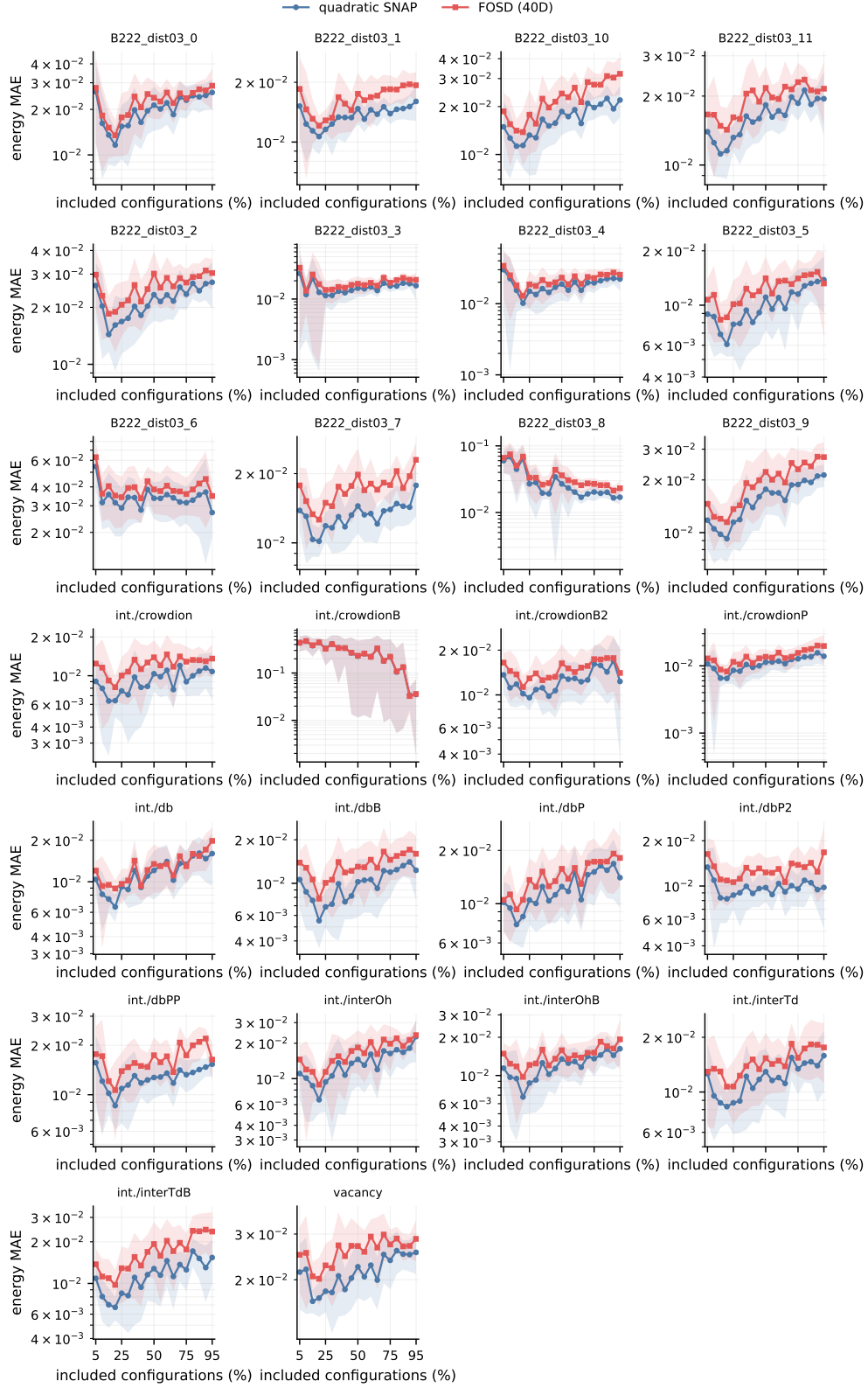


Figure 6: Re held-out energy MAE for all 26 source-defined groups.

C Sensitivity and architecture attempts

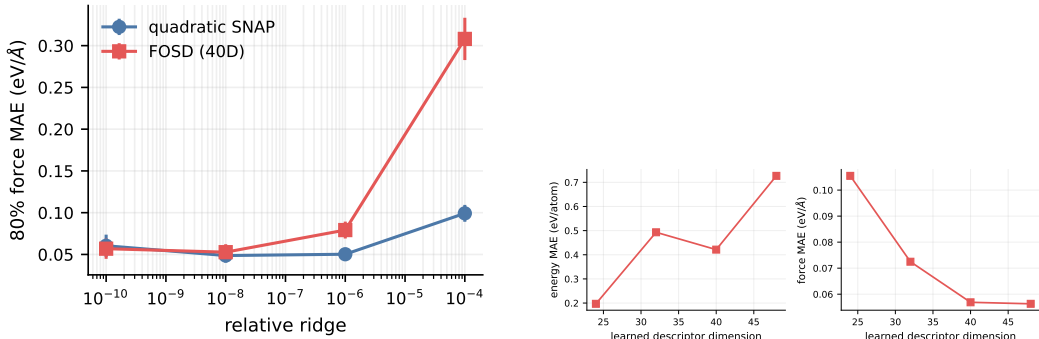


Figure 7: Post-freeze Ta sensitivity at 80% included data, always using all ten draws. Left: force MAE versus final-head ridge. Right: FOSD energy and force MAE versus output dimension. Complete 50%, 80%, 95%, included/excluded, and groupwise cells are retained in the ablation logs.

Table 6: Actually run Ta architecture attempts on the fixed 80%, draw-0 split. Full errors use all included configurations. All heads are the required linear-plus-quadratic least-squares form.

Descriptor bank	Dim.	Cutoff (Å)	Held-out δE	Held-out δF	Full $\delta E / \delta F$
31D POD two-/three-body	24	5.0	0.06295	0.4705	0.03129 / 0.3790
72D POD through four-body	32	5.2	0.05127	0.5716	0.02232 / 0.3411
72D POD + 128 fixed $\tanh$, Jacobian distillation	40	5.2	0.117015	1.66685	0.017279 / 0.299795
Final bispectrum FOSD	40	5.594	0.008356	0.059588	0.000438 / 0.020836

D Audit accounting

The raw CV metric files contain 10,668 Ta rows, 15,240 Mo rows, 16,002 Nb rows, and 20,574 Re rows after including method, partition, material group, and aggregate cells. For each material the validator finds exactly 38 excluded aggregate method–fraction cells, ten unique draws in every cell, two full endpoints, complete source-group panels, no duplicated method–fraction–draw–partition–group keys, and no missing values. The four invariance logs contain 9,600 transformed predictions: 2,400 per material and 1,200 per model within material. The four timing logs contain 40 runs: ten per material and five per model. These counts are generated from the saved files rather than inferred from figure markers.